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Journal and Proceedings, Royal Society of New South Wales, Vol. 105, pp. 86–93, 1972

The Palaeomagnetism of Some Palaeozoic Sediments from Central Australia

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Communicated by DR. K. A. W. CROOK

ABSTRACT—Stable components of magnetic remanence have been isolated (using the partial thermal demagnetization technique) in rock samples collected from the Middle Ordovician Stairway Sandstone and the Siluro ?–Devonian Mereenie Sandstone. The stable magnetic remanence is interpreted to be of primary origin. The Stairway Sandstone, sampled over about 350 metres, yields a palaeomagnetic south pole situated at 2°S, 50·5°E ($\alpha_{95}=8·5°$) and the Mereenie Sandstone sampled over the lower 100 metres of a 300 metre succession, yields the pole position 41·5°S, 40·5°E ($\alpha_{95}=10·5°$). The Ordovician pole supports previous results but the Siluro ?–Devonian pole is situated on an older portion of the Australian polar wander curve than would be expected when compared with published Silurian data.

1. Introduction

This paper deals with the palaeomagnetism of the Stairway Sandstone (of Middle Ordovician age) and the Mereenie Sandstone (at least in part Devonian but possibly extending back into the Silurian) and complements previous work reported by Embleton (1972). Since the early investigations by Irving and Green (1958), few results have been obtained from Lower Palaeozoic rock formations within Australia. The polar wander curve for Australia, beyond the Carboniferous, is based on Cambrian data from the Northern Territory (McElhinny and Luck, 1970 and Embleton, 1972), a single Ordovician

TABLE 1a

Stairway Sandstone : summary of directions of remanent magnetization with respect to bedding, before cleaning (NRM) and after treatment (600–630°C).

	NRM					600–630°C			
Sample	N	R	D°	I°	Int. × 10^{-6} Gauss	N	R	D°	I°
A70	2	1·932	297·5	+15·5	10·7	2	1·941	259·0	+20·0
A71	3	2·962	327·0	+16·5	15·1	3	2·961	268·5	+14·0
A72	3	2·999	273·5	+22·5	7·3	3	2·998	277·0	+23·0
A73	4	3·966	299·0	+28·5	9·8	3	2·970	259·5	−0·5
A74	3	2·970	282·0	+8·0	12·2	2	1·998	276·5	+11·0
A75	3	2·999	297·0	+13·0	5·6	3	2·960	274·5	+16·0
A76	2	1·999	308·5	+20·5	15·6	2	1·990	281·0	+34·0
A77	4	3·989	316·0	+30·5	16·4	3	2·707	266·0	+14·5
A78	3	2·956	292·5	+25·0	10·7	3	2·904	251·0	+5·0
A79	3	2·990	324·5	+17·5	58·7	2	1·965	298·0	+6·0
A80*	3	2·999	353·0	−5·5	86·3 †		—	349·5	−1·5
A81*	2	1·999	360·0	+5·5	85·2 †		—	358·5	+7·5
A82*	3	2·999	358·0	−4·0	75·3 †		—	6·0	+9·0
A83*	3	2·990	350·0	+4·5	98·5 †		—	348·0	+12·5

* Samples " remagnetized ", contain a thermally stable secondary component.
† Pilot specimens.

TABLE 1b

Formation mean direction and pole position after thermal treatment.

	n	R	D°	I°	$\alpha°_{95}$	Lat.	Long.	$\alpha°_{95}$
(i) 600–630°C	10	9·636	271·0	+14·5	10·0	2°S	50·5°E	8·5
(ii) 630–660°C	10	8·714	275·0	+22·5	19·5	0°	59·0°E	16·0

Symbol notation : N=number of specimens, n, number of samples, R=resultant of N or n unit vectors, D°=declination, I°=inclination, Int.=intensity of NRM, α_{95}=semi-angle of the cone of confidence at the 95% probability level.

result from the Northern Territory (Luck, 1970) and a result for the Silurian from eastern New South Wales (Briden, 1966). Several recent interpretations of the data can be seen in McElhinny and Luck (1970), Creer (1970) and Embleton (1972).

about 60 percent of fine and medium grained quartz grey-wacke, sometimes quite silty and about 40 percent of cleaner quartz sandstone. Ten oriented samples (A70–A79) were collected from five deep red siltstone bands, ($\frac{1}{2}$ metre–$1\frac{1}{2}$ metres thick) which occur at intervals

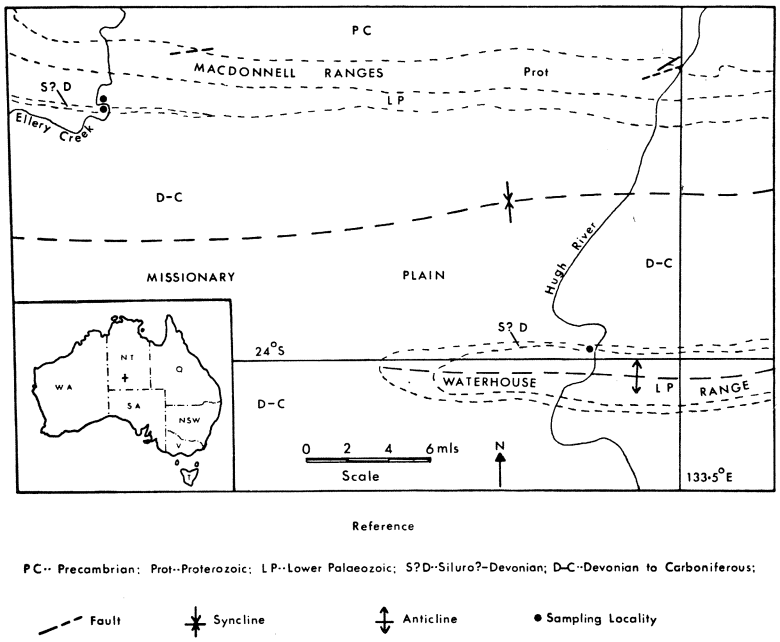


FIGURE 1.—Geological sketch map of the sampling locality. Dots indicate the localities from which samples were obtained.

Localities in which to sample material for palaeomagnetic investigation were chosen remote from south-east Australia in view of the complex geological history that region suffered (Brown *et al.*, 1968), and following some modern ideas which question the relative geological antiquity of south-east Australia and the physical integrity of the region during the Lower Palaeozoic (Oversby, 1971).

2. Brief Geology of the Area Sampled

The Stairway Sandstone and the Mereenie Sandstone occur as successive stratigraphic units in the Ellery Creek section of the Palaeozoic sedimentary sequence in the Amadeus Basin, central Australia (23·8°S, 133°E). Palaeomagnetic results have been described from the Arumbera Sandstone (Proterozoic-Cambrian) and the Hugh River Shale (Lower-Middle Cambrian) from the same locality by Embleton (1972). According to Prichard and Quinlan (1962) the Stairway Sandstone is about 350 metres thick at Ellery Creek and consists of throughout the middle and upper beds of the sequence. Four samples (A80–A83) of fine-grained sandstone, rather pinkish and poorly cemented, were collected near the top of the formation. The beds dip uniformly southwards at about 60°. The age of the Stairway Sandstone is well established palaeontologically (Wells *et al.*, 1970), and ranges from Upper Llanvirnian to Llandeilian.

The Mereenie Sandstone unconformably overlies the Stairway Sandstone. The beds again dip uniformly southwards but with a somewhat shallower inclination—about 55°. Twenty-eight oriented samples (A84–A112, A93 was not collected) were collected from the lower beds where it forms a prominent " . . . red wall on the north side of the curve in the Creek (Ellery) just before the stream enters the congolmerate hills (Pertnjara Group) " (Madigan, 1932, p. 690). In the sampling locality, the formation is about 300 metres thick, and approximately the lower 100 metres were sampled for palaeomagnetic study. It is a fine to medium grained, red-brown

sandstone with cross-bedding more conspicuously developed in the middle and upper beds of the sequence. Few fossils have been found in the Mereenie Sandstone, but Milton (1967) has reported the presence of some Devonian fish fragments, so that it is in part Devonian. The overlying Parke Siltstone (basal member of the Partnjara Group) is also Devonian in age (Wells *et al.*, 1970). In sections through the Palaeozoic sequence further west, the Mereenie Sandstone apparently rests conformably on the Carmichael Sandstone. The latter formation is considered Upper Ordovician in age. It therefore seems probable that the Mereenie Sandstone could extend back into the Silurian—hence the designated age of Siluro ?–Devonian—but as yet there is no fossil evidence to support this. A further five samples (A144–A148) were collected from north-dipping beds in the Waterhouse Range about twenty-five miles south-east of Ellery Creek. A geological sketch map of the sampling locality is shown in Figure 1.

3. Palaeomagnetic Results

Cylindrical specimens (2·8 cm. dia., 2·8 cm. high) were cut from the oriented samples and measured for directions and intensities of magnetization using a high sensitivity spinner magnetometer. The partial thermal demagnetization technique was employed to (i) test the stability of the magnetic remanence and (ii) remove secondary components of magnetization. Directions of magnetization were analysed following the method devised by Fisher (1953) and palaeomagnetic pole positions were calculated by assigning unit weight to each sample. In Figures 2, 4 and 5, the directions are plotted stereographically on a Wulff net.

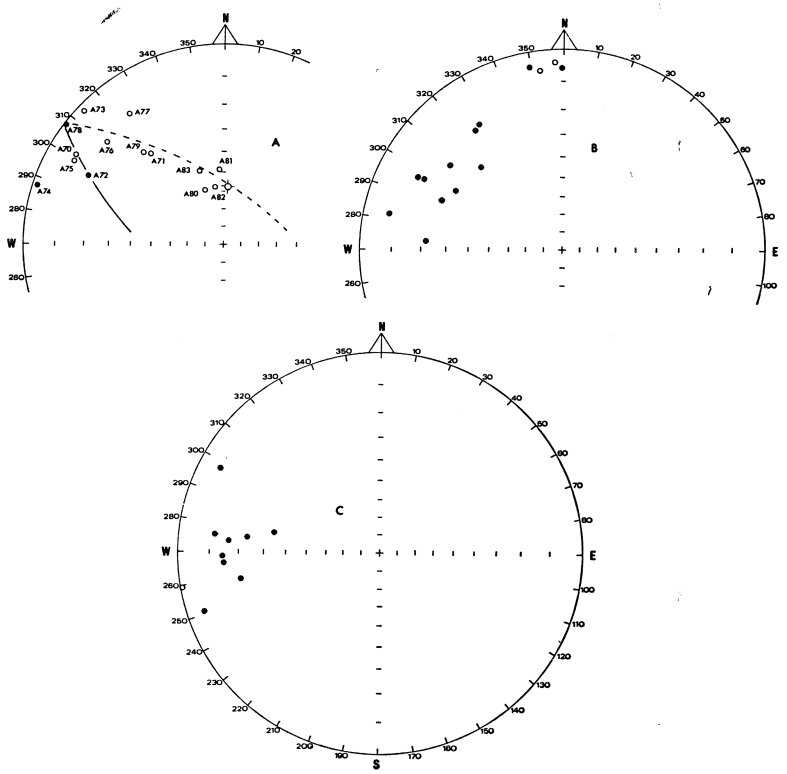


FIGURE 2.—Sample-mean directions of remanent magnetization measured in the Stairway Sandstone. Parts A and B show the NRM directions referred to the present horizontal and palaeohorizontal (bedding plane) respectively; the NRM distribution is streaked through the present field direction denoted by the open star in A. Part C shows the 600–630° C population; the streaking has been eliminated. Open symbols represent negative (upward pointing) directions and closed symbols positive (down pointing) directions.

3.1 The Stairway Sandstone

Sample-mean directions of natural remanent magnetization (NRM) exhibit a streaked distribution which, when plotted with respect to the present horizontal, are approximately directed along a great circle passing through the present field axis (see Figure 2A). The four samples that were collected near the top of the formation (numbers A80–A83) group close to the direction of the present field, and their NRM intensities (Table 1a) are also much higher than the NRM intensities of samples A70–A78. Although sample A79 also had a high NRM intensity, it did respond to thermal treatment—whereas samples A80–A83 maintained their initial directions of magnetization up to temperatures in the region of 650° (Table 1a lists their directions up to 630°C and since no apparent change has occurred, they are classed as " remagnetized "). Streaking was almost eliminated from the remainder of the samples after partial thermal demagnetization at about 500°C. They were treated at two further steps to effectively isolate the primary component of magnetic remanence. Sample-mean data after treatment at 600–630°C is listed in Table 1a and Table 1b compares the 600–630°C sample population with the 630–660°C population. The demagnetization-intensity curves shown in Figure 3 illustrate the relatively lower stability of magnetic remanence of samples A80–A83 compared with samples A70–A79, although some overlap in stability is apparent. The isolated component of remanence (reversed polarization, Figure 2c) is strongly oblique to the present geomagnetic field direction and is quite stable up to the blocking temperatures of magnetic mineral phases (probably in the region 600–650°C). The position of the south palaeomagnetic pole computed from sample virtual geomagnetic poles (VGPs), after treatment at 600–630°C, is 2°S, 50·5°E (α_{95}=8·5°).

3.2 The Mereenie Sandstone

Relatively strong secondary components of magnetic remanence have been acquired in the direction of the present geomagnetic field—Figure 4A shows the thirty-three sample-mean directions tightly grouped about the present field axis (situated at 005°, −58°). After correction for bedding tilt two populations emerge: (i) comprised of samples from south-dipping beds (A84–A112), and (ii) of samples from the north-dipping beds (A144–A148). None of the samples from the latter group responded successfully to thermal treatment, so precluding the possibility of applying the fold test (Graham, 1949) after treatment. Eleven of the twenty-eight samples from the south-dipping beds were successfully " cleaned ". Fourteen of the remainder (samples A84–A88 and A95–A103) maintained their NRM directions up to 650°C, and the three samples A91–A94 behaved erratically. Sample-mean intensities of remanent magnetization varied in such a way that those samples which ultimately responded

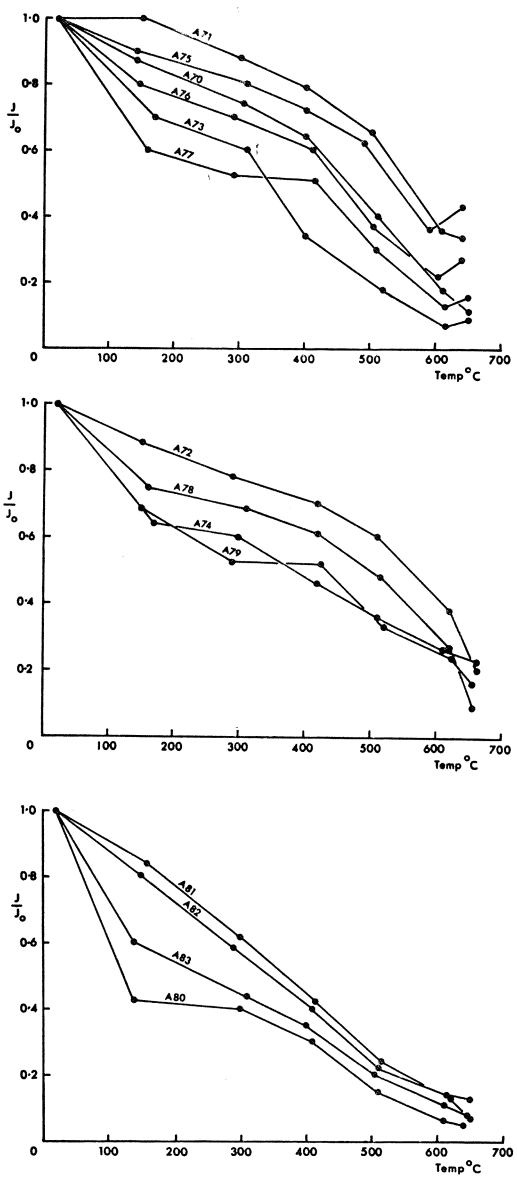


FIGURE 3.—Normalized demagnetization-intensity curves for pilot specimens of Stairway Sandstone. (J/J_0 is the ratio of the intensity of magnetization at a particular temperature to the NRM intensity value).

successfully to treatment had the lowest NRM values. The arithmetic-mean NRM values of intensity for each group were $2{\cdot}8\times10^{-5}$ gauss (accepted samples, $n=11$) and $1{\cdot}1\times10^{-4}$ gauss (rejected samples, $n=22$).

Pilot specimen data from the accepted samples is shown in Figure 5—only after high-temperature treatment did they reveal any tendency to group significantly away from the present field axis. Further specimens were then treated at 590–620°C and at 620–650°C. Figures 4C and D indicate the reduced scatter achieved at the higher of the two temperatures. Specimen directions from two samples (A89 and A109) rotated into a position reversed with respect to the main population—they were not collected from a common stratigraphic horizon so the implication of the reversal remains obscure. However, the presence of both polarities, approximately anti-parallel and oblique to the present field axis, lends weight to the significance of the result even though the origin of the phenomenon is uncertain. Table 2 lists the sample-mean data after "cleaning" and the corresponding palaeomagnetic south-pole calculated from sample VGPs is situated at 41·5°S, 40·5°E ($\alpha_{95}=10{\cdot}5°$).

4. Discussion

Interpretation of the results reported here will be restricted to the implications they have for the existing palaeomagnetic record of Australia. All samples of Stairway Sandstone which were judged to have retained a proportion

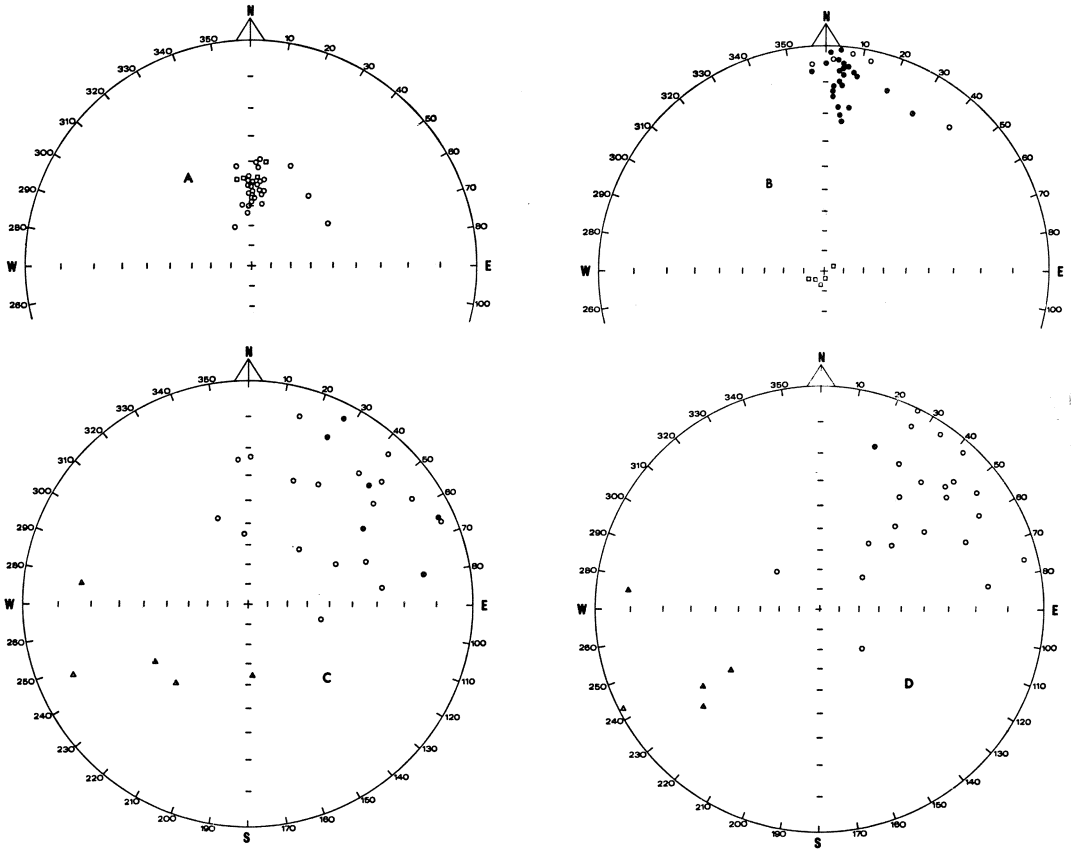


FIGURE 4.—Directions of remanent magnetization measured in the Mereenie Sandstone. Parts A and B show NRM sample-mean directions referred to the present horizontal and the palaeohorizontal respectively. The cluster around the present field axis at about 005°, −58° (part A) indicates that the bulk of the remanence is of recent origin—this is fully substantiated by the two groups of directions revealed after applying the bedding correction (part B); the folding is late Palaeozoic. Samples collected from the south-dipping limb are denoted by circles (A84–A112) and those from the north-dipping limb by squares (A144–A148). Parts C and D show the specimen populations after treatment at 590–620° C and 620–650° C respectively. Triangles denote specimens with reversed polarity.

of their primary remanence were reversely magnetized. The computed mean pole position agrees reasonably well with the Lower Ordovician pole position reported by Luck (1970). The Jinduckin Formation, sampled at locality 14°S, 132°E, yielded a palaeomagnetic south pole position at 16°S, 29°E (α_{95}=17°). Both poles are shown in Figure 6 and lie close to the apparent polar wander curve connecting the Lower-Middle Cambrian and the Siluro ?–Devonian polar regions. No other investigations have successfully produced palaeomagnetic poles from rocks of Ordovician age in Australia.

TABLE 2

The Mereenie Sandstone, summary of sample-mean data after partial thermal demagnetization. Directions are restored to the bedding planes.

	590–620°C				620–650°C			
Sample	N	R	D°	I°	N	R	D°	I°
A89 ..	2	1·513	243·5	+47·5	2	1·870	257·0	+18·5
A90 ..	2	1·949	81·5	−50·5	2	1·970	80·5	−60·0
A104 ..	2	1·984	61·0	−2·0	2	1·997	56·0	−9·5
A105 ..	2	1·805	33·0	−6·5	2	1·898	41·5	−32·0
A106 ..	3	2·737	31·0	−17·5	3	2·869	42·5	−14·5
A107 ..	3	2·338	28·0	−27·0	3	2·678	37·0	−18·5
A108 ..	3	2·727	44·5	−4·0	3	2·809	24·5	−35·0
A109 ..	3	2·728	237·5	+23·0	3	2·867	236·0	+20·0
A110 ..	3	2·621	54·0	+0·5	3	2·818	51·0	−13·0
A111 ..	2	1·891	41·0	−34·0	2	1·789	48·2	−37·5
A112 ..	3	2·472	35·0	−18·5	3	2·714	46·5	−7·0
Mean n=11		10·226	47·0	−21·5	n=11	10·353	50·0	−24·5

Mean of sample VGPs after 620–650°C treatment: (samples A89 and A109 reversed)

n	R	Lat.	Long.	α_{95}
11	10·474	41·5°S	40·5°E	10·5°

Symbol notation as for Table 1.

Much palaeomagnetic work has been carried out on Devonian rock formations, but the absence of laboratory stability tests has led to the results being omitted from the polar-wander curve (for example see Creer (1970) and McElhinny and Luck (1970)). Results from two rock sequences which were " cleaned " (viz. the Housetop Granite plus aureole (Briden, 1967) and the Dotswood red beds (Chamalaun, 1968)) indicated that their magnetic remanence is younger than the rock ages, so they also have been omitted. There are therefore no results with which to compare directly the pole yielded by the Mereenie Sandstone. A Silurian pole position is available from the Mugga Mugga Porphyry, reported by Briden (1966), but the pole obtained from the Mereenie Sandstone, can be seen from Figure 6 to lie on an *older* portion of the apparent polar-wander curve. This discrepancy may be explained by one of the following possibilities: (i) one or the other, or even both rock units have not retained their primary remanence, (ii) the rock ages are incorrect, or (iii) relative displacements of the crustal regions containing each rock unit have occurred since their formation, either by local block rotation or by translation. With regard to possibility (i), the directions of magnetic remanence are believed to be primary in both cases, but a further point is whether the secular variation of the geomagnetic field has been accounted for by sampling over a sufficient time interval. Explanation (ii) certainly does not appear as a likely solution—the physical age of the Mugga Mugga Porphyry, determined using the Rb-Sr isochron method (Bofinger *et al.*, 1970) is 423±9 My (^{87}Rb decay constant $1{\cdot}39\times10^{-11}$ yr^{-1}) and the Devonian age for the Mereenie Sandstone is based on the presence of diagnostic fish fragments which occur in the sequence (see section 2). Furthermore, there is no suggestion that the Sandstone could be older than Silurian. The Mugga Mugga Porphyry is located in south-east Australia, a region whose suitability as a collecting ground for Lower Palaeozoic material has been questioned by the author on the basis of arguments which raise doubt about the physical integrity of south-east Australia at that time. It may be that possibility (iii) has contributed to the disparity between the results. It is unlikely that central Australia has suffered significant translational or rotational movements relative to the main shield areas, at least since the uppermost Pre-Cambrian (Embleton, 1972). Until further investigations are made, however, it remains premature to reach a definite conclusion

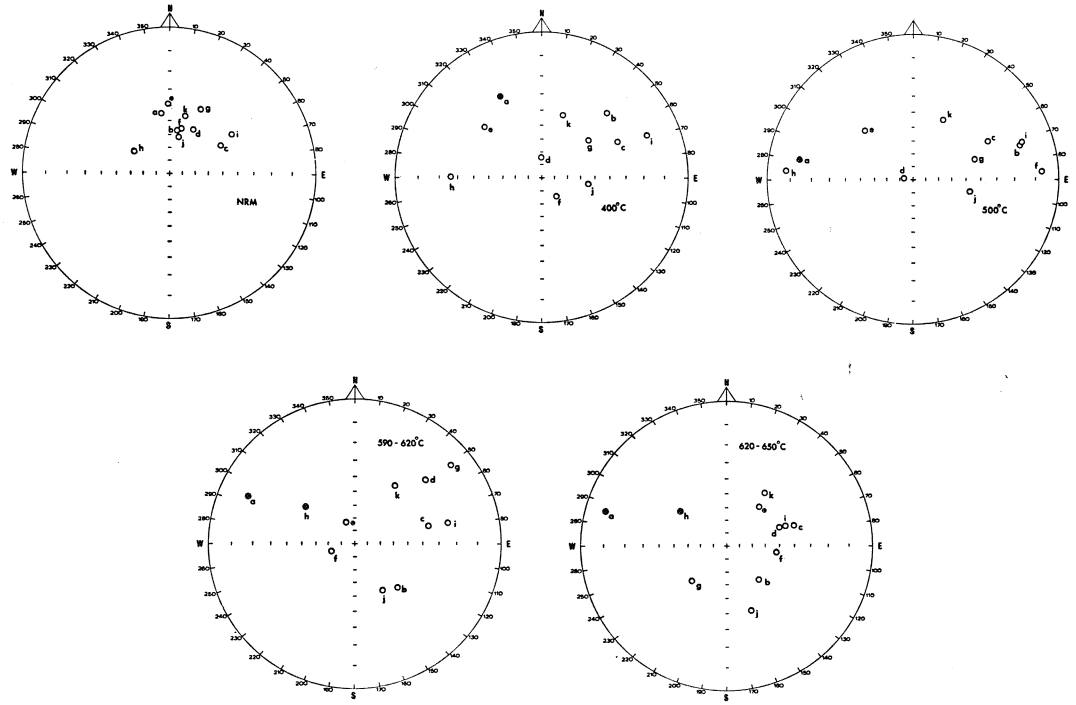


FIGURE 5.—Mereenie Sandstone pilot specimen populations; directions are referred to the present horizontal. A letter is allocated to each specimen for comparison of its direction after treatment at the various temperatures indicated in the Figure.

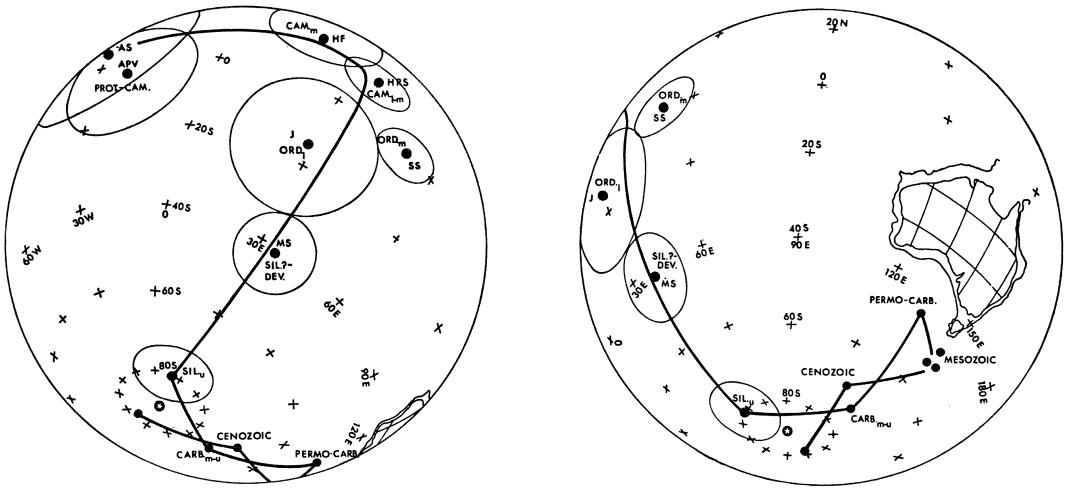


FIGURE 6.—The polar wander curve for Australia. All Lower Palaeozoic pole positions are plotted with their respective circles of confidence at the 95 per cent probability level (P=0·05), Fisher (1953). APV=Antrim Plateau Volcanics, McElhinny and Luck (1970); AS=Arumbera Sandstone and HRS=High River Shale, Embleton (1972); HF=Hudson Formation, Luck (1972); J=Jinduckin Formation, Luck (1970); SS=Stairway Sandstone and MS=Mereenie Sandstone, this paper. The Upper Silurian pole position was reported by Briden (1966) from a study of the Mugga Mugga Porphyry. The Upper Palaeozoic and younger portion of the curve has been described by Irving (1966) and McElhinny and Luck (1970).

about the Silurian and Devonian palaeomagnetic pole position which is valid for the main shield area of Australia.

Acknowledgement

The author expresses his thanks to Mr. D. J. Edwards for the competent assistance he offered in the field.

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(Received 26 June 1972)

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